Automatic Modulation Classification Engine Using 1D Convolutional Neural Networks

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Date: August 11, 2026

Abstract

Automatic Modulation Classification (AMC) plays a critical role in non-cooperative communications, spectrum monitoring and modern Software-Defined Radio (SDR) architectures. Traditional likelihood-based and feature-based AMC methods suffer from high computational complexity and require manual domain-specific feature extraction. This project proposes an end-to-end deep learning framework utilizing a 1D Convolutional Neural Network (1D CNN) to classify modulation schemes directly from raw complex In-Phase and Quadrature (I/Q) time-series samples. The proposed system aims to classify 11 analog and digital modulation schemes across a wide range of Signal-to-Noise Ratios (SNR) with low-latency inference suitable for edge deployment.

1. Introduction

Modern wireless environments are increasingly crowded and dynamic. Cognitive Radios (CR) and automated spectrum surveillance systems require real-time identification of incoming signals without prior knowledge of transmitted data or signal parameters.

Automatic Modulation Classification (AMC) serves as an intermediate step between signal detection and demodulation. While 2D CNNs have gained popularity by treating spectrograms or constellation diagrams as images, they introduce additional computational overhead during the signal-to-image transformation process. A **1D Convolutional Neural Network (1D CNN)** directly processes raw complex baseband representations, reducing the computational footprint while preserving intrinsic phase and amplitude trajectories.

2. Project Objectives

The primary objectives of this project are:

1. Develop an end-to-end deep learning pipeline to classify 11 modulation schemes using the RadioML 2016.10A benchmark dataset.
2. Implement a lightweight 1D CNN architecture capable of processing $2 \times N$ raw complex I/Q tensor inputs.
3. Evaluate classification accuracy across varying SNR levels ranging from $-20\,\mathrm{dB}$ to $+18\,\mathrm{dB}$.
4. Optimize the trained model using INT8 quantization to achieve low-latency inference suitable for embedded edge targets.

3. Theoretical Background

3.1. Complex Baseband Representation and I/Q Signals

A bandpass Radio Frequency (RF) signal $s(t)$ centered at carrier frequency f_c can be expressed as

$$s(t) = A(t) \cos \left(2\pi f_c t + \phi(t)\right), \tag{1}$$

where $A(t)$ is the instantaneous envelope and $\phi(t)$ is the instantaneous phase.
The corresponding complex baseband representation can be written as

$$r(t) = I(t) + jQ(t), \tag{2}$$

where the **In-Phase (I)** and **Quadrature (Q)** components are defined as

$$I(t) = A(t) \cos \left(\phi(t)\right), \tag{3}$$
$$Q(t) = A(t) \sin \left(\phi(t)\right). \tag{4}$$

In a digital sampling system, the complex samples can be arranged into a two-channel 1D input matrix $\mathbf{X} \in \mathbb{R}^{2\times N}$ over N sampling intervals:

$$\mathbf{X} = \begin{bmatrix} I_1 & I_2 & \dots & I_N \, Q_1 & Q_2 & \dots & Q_N \end{bmatrix}. \tag{5}$$

3.2. Modulation Schemes and Constellations

Digital modulation schemes encode information through controlled changes in the phase, amplitude, or frequency of a carrier signal. The principal modulation categories considered in this project include:

- **Phase Shift Keying (BPSK, QPSK, 8PSK):** Information is primarily represented through phase transitions $\Delta\phi$, with an approximately constant envelope magnitude $|r(t)|$.
- **Quadrature Amplitude Modulation (16-QAM, 64-QAM):** Information is represented through simultaneous variations in amplitude and phase, producing a structured two-dimensional constellation.
- **Frequency Shift Keying (CPFSK, GFSK):** Information is encoded through controlled variations in the instantaneous frequency of the signal.

3.3. 1D Convolutional Neural Networks for Time-Series Signals

Unlike spatial 2D convolutions, a 1D convolution slides a filter kernel w of length K along the temporal axis of a time-series input.

For an input sequence x, the convolution operation at time index t for output channel c can be expressed as

$$y_c[t] = \sigma \left(\sum_{m=1}^{C_{\text{in}}} \sum_{k=0}^{K-1} w_{c,m}[k], x_m[t-k] + b_c \right), \tag{6}$$

where C_{in} represents the number of input channels, $w_{c,m}$ denotes the convolutional filter weights, b_c is the bias term and $\sigma(\cdot)$ represents a nonlinear activation function such as the Rectified Linear Unit (ReLU).

For the proposed AMC system, the two input channels correspond to the In-Phase (I) and Quadrature (Q) components.

4. Methodology & System Architecture

4.1. Dataset & Preprocessing

The model will be trained using the **RadioML 2016.10A** benchmark dataset, which contains signal examples from 11 modulation classes over a range of SNR conditions.

Each sample is energy-normalized to approximately unit average power. This normalization reduces the dependence of the network on absolute signal amplitude and encourages the model to learn modulation-specific characteristics.

The normalized input is calculated as

$$\hat{\mathbf{X}} = \frac{\mathbf{X}}{\sqrt{\frac{1}{N}\sum_{k=1}^{N}\left(I_k^2 + Q_k^2\right)}}. \tag{7}$$

Here, $\hat{\mathbf{X}}$ denotes the normalized I/Q sample matrix.

4.2. Proposed 1D CNN Model Architecture

The proposed network consists of three 1D convolutional feature extraction blocks followed by adaptive global average pooling and a fully connected classification head.

Table 1: Proposed 1D CNN Architecture Specification

Layer Type	**Kernel**	**Stride**	**Output Shape**
Input Tensor	–	–	$(2, 128)$
Conv1D + BN + ReLU	7	1	$(64, 128)$
MaxPool1D	2	2	$(64, 64)$
Conv1D + BN + ReLU	5	1	$(128, 64)$
MaxPool1D	2	2	$(128, 32)$
Conv1D + BN + ReLU	3	1	$(256, 32)$
AdaptiveAvgPool1D	–	–	$(256, 1)$
Dense + Dropout (0.5)	–	–	(128)
Dense Output	–	–	(11)

5. Expected Deliverables

The expected outcomes of the project are:

1. A well-structured PyTorch code repository containing modular preprocessing, dataset-loading, training and evaluation routines.

2. Performance curves demonstrating the classification performance of the proposed model across different SNR conditions, with a target accuracy exceeding 90% at high SNR levels (SNR $\geq 0\,\mathrm{dB}$).

3. A quantized ONNX model optimized for low-latency inference and potential edge deployment.

4. A complete technical project report documenting the theoretical background, methodology, implementation, experimental results and conclusions.

References

[1] T. O'Shea and J. Hoydis, "An Introduction to Deep Learning for the Physical Layer," *IEEE Transactions on Cognitive Communications and Networking*, vol. 3, no. 4, pp. 563–575, 2017.

[2] T. J. O'Shea, J. Tamir, N. Clancy and T. C. Clancy, "Over-the-Air Deep Learning Based Radio Signal Classification," *IEEE Journal of Selected Topics in Signal Processing*, vol. 12, no. 1, pp. 168–179, 2018.